

USG Notes

Golden Triangle

- **What is our Golden Triangle?**
 - Example: Boss Fight (Pick 2)
 - Tough (High HP)
 - Fast (Speedy)
 - Strong (High Damage)
 - Possible Planet Pillars
 - Danger (High Damage, Radiation, Inclement Weather, Lava Pools, etc.)
 - Durability (High HP, Larger Planets, Impassible Materials)
 - Value (Lucrivity, Scarcity, Technology)
 - Element Pillars
 - Weight
 - Durability
 - Value
 - Tetherable vs Collectable

Movement

- **System View**
 - Illusion of 6 degrees of movement
 - Really no up / down
 - *Cruise Mode*
 - Player flies around above planets
 - Planets grow in scale depending on player proximity
 - *Intercept Mode*
 - Player flies on same plane as planets
 - Gravity is in effect, allowing player to slingshot around planets

Game Aesthetics

- Chunky
- Blocky
- Geometric
- Mobile games meets Straylight

Mini games. Landing. Orbiting.

Mining: Harvest rare elements and minerals

Salvage:

Archeology:

Teleport down to the planet from orbit.

Mini game: warp between systems. Must stay inside the warp pipe or be thrown out early.

- This could be how ships wind up between systems.

Next Todo:

- Focus System
 - Pressing spacebar will cycle through available targets in range
 - When player is locked on:
 - If player velocity is close to target velocity, match player velocity to target velocity
 - UI: return information about the target object
 - Activate gravity gauge (Line Renderer) for satellite objects or non-landable objects
 - Activate Altimeter for landable objects

Todo:

- Animate thruster with random length and rotation when forward key is pressed
- Swap Camera between ship and orbited planet
- Galaxy Map / Warp System
- Bind T key to Target nearest planet:
 - Match movement
 - Gain information
 - Mass
 - Gravity
 - Atmo
 - Radius
 - AU
 - Lock Camera onto Planet
 - Adjust Dolly param based on distance
 - Activate Orbital Attitude Line Renderer
- Waypoint markers to point off screen towards objectives / POIs

Orbital Attitude LR

- Child of ship
- Fires forward to track ship's potential trajectory
- Used to safely land or orbit a planet

***** Restructure Notes *****

- Player class will contain all eventual player-related information
- Spatial Bodies will generate a Surface Collider based on scale.
- SB's will have a gravity field Collider that will act as a trigger for applying gravitational forces to Orbital objects
- public interface Orbital
 - Require the necessary internals to allow an object to be affected by a gravity field
- public interface Atmosphere
 - bool hasAtmo = true;
 - Atmosphere Collider to add drag around planet
- Planetary gravity should be handled by its own script

- this will promote readability and modularity

Parents : Children

- I. SpatialBody
 - A. Sun
 - 1.
 - B. Planet
 - 1. Rocky
 - 2. Ice Giant
 - 3. Gas Giant
 - C. Satellite
 - 1. Moon
 - 2. Station

Notes on Controller vs Handler:

- Controller tends to be broader in spectrum, managing the overall flow of a scene, level, or gameplay mechanic
 - ex. LevelController - manage level progression, spawning enemies, loading new areas, etc.
- Handler is more focused, handling specific events or data types
 - ex. Input Handler captures player input and translates it into movement or actions, or
 - CollisionHandler handles collision events between objects; triggers appropriate responses

Game Design Document

Orebit (???)

Game Design Document

Overview

In this roguelite space-mining survival game, players choose from three unique planets per run to mine resources, battle enemies, and survive intense conditions. Core gameplay revolves around exploration, mining, and strategic combat using passive defense mechanics inspired by Brotato and Vampire Survivor.

Gameplay Mechanics

Planet Selection

- **Standard Planets:**
 - In each run, players select 1 of 3 procedurally generated planets.
 - Distinct environmental characteristics (lava flows, ice caves, radioactive zones).
 - Used for resource mining, enemy encounters, and leveling up permanent upgrades.
 - Planets have 3 levels. Burrow Directly to L2 or L3. Balatro style skip blinds. Fewer resources gathered so it's harder, but bigger reward, less time wasted.
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- **Boss Planets:**
 - Four permanent planets containing ship components essential for repairing the Arc.
 - Persistent tunnels across multiple visits; incremental progress allowed.
 - Higher difficulty: tougher terrain, stronger enemies proportional to the planet's level.

Player Entry and Movement

- Players directly drop onto planet surfaces using:
 - Small landing craft (non-pilotable)
 - Personal space suit equipped with a jetpack (to be finalized).
- Drop ship upgrade technology
 - 10 Pt system. (10% further dig)

Exploration and Mining

- Worlds feature procedurally generated terrain including destructible landscapes and dynamically created underground tunnels:
 - Virgin soil: randomly generated free-form areas that ensure diverse and unique mining experiences every run.
 - Pre-dug tunnels: algorithmically placed structures featuring a combination of linear and branching paths, enemy encounter zones, and specialized rooms such as treasure vaults.
- Mining:
 - Resource gathering to craft upgrades and defenses.
 - Strategic mining is required to manage fuel, oxygen, and resource capacity.

Combat and Defense System

Enemies

- **Ant-like creatures:** Basic threat, moderate speed, and damage.
- **Flying creatures:** Fast, darting creatures that follow you regardless of terrain
- **Beetle-like creatures:** Armored; slow-moving but heavy damage resistance.
- **Elemental Monsters:** Adapt to planet type (lava, ice, radioactive, toxic); special abilities, and area effects
- **Mammals:** Organic burrows and tunnels. Larger boss/treasure rooms. Harder more advanced tech

Rooms and Encounters

- Larger rooms containing:
 - Rich resource deposits.
 - Miniboss encounters provide significant upgrade rewards.

Passive Defense System

- Players possess a suite of automatically firing weapons and abilities to handle enemy hordes.
- Defensive upgrades include:
 - Orbiting drones and turrets.
 - Reactive armor and environmental shields.
 - Area control abilities (elemental fields, slowing zones, damage-over-time effects).

Progression and Upgrades

- Permanent upgrades are unlocked through mining, defeating minibosses, and surviving planetary challenges.
- Upgrades enhance:
 - Mining efficiency.
 - Jetpack movement and fuel efficiency.
 - Passive defense capabilities (increased fire rate, damage, additional defensive units).

Visual and Audio Style

- Atmospheric visuals tailored to planet themes.
- Bangin' follow-up soundtrack to the award-winning STRAYLIGHT OST
- Rich, ambient soundscapes enhance immersion and tension.
- Retro-inspired UI and aesthetics with modern fluid animations.

This document will serve as the foundational guide for further detailing individual systems and facilitating development planning.

Animal Cast

Classes / Skills

- Capybara - the Tank, focus Survival
 - Beast Mode - Become temporarily invulnerable
- Beaver - the Engineer, focus Resilience
 - Hard Reboot - Turn your ship off and on again, resetting health and life support
- Fox - the Ace, focus Navigation
 - Outta Here - Jump to the surface of a planet
- Prairie Dog - the Scout, focus Digging
 - Deep Focus - Instantly scan all bodies w/in range
- Squirrel - the Hoarder, focus Resource Gathering
 - Rummage Sale - Increased odds of finding rare or valuable materials
- Chipmunk - the Speedster, focus Exploration
 - Phase Shift - Become temporarily incorporeal

Animal	Size	Exp	RG	Sur	Focus
Capybara	Very Large	1	1	3	Survival
Beaver	Large	1	2	2	Resilience
Fox	Medium	2	1	2	Navigation
Prairie Dog	Small	2	2	1	Digging
Squirrel	Very Small	1	3	1	Resource Gathering
Chipmunk	Tiny	3	1	1	Exploration

Traits:

- Exploration
 - Discover and Navigate the Cosmos
 - Involves
 - Planetary Scanning
 - Valuable resources
 - potential hazards
 - POIs
 - Navigation
 - Charting courses
 - Optimizing Travel Routes
 - Avoiding Dangers / Hazards
 - System Mapping
 - Discover new star systems
 - Charting system layouts
 - Identifying potential jump points
 - Anomaly Detection

- Finding and investigating unusual phenomena
 - Locating missing ship parts / lost crew
- Resource Gathering
 - Acquiring materials necessary for survival and advancement
 - Involves
 - Mining: Extracting resources from planets & asteroids
 - Salvaging: Recovering valuable materials from derelict ships / space debris
 - Resource Management: Optimizing storage, usage, refinement of materials
- Survival
 - Ensuring the continued operation of the ship and crew in the face of dangers
 - Ship Maintenance
 - Repairing damage
 - Upgrading systems
 - ensuring optimal ship performance
 - Environmental Protections: Shielding the ship from
 - Radiation
 - Extreme Temperatures
 - Environmental Hazards
 - Life Support Mangement
 - Maintaining Oxygen Levels
 - Maintaining Life Support Systems
 - Med Bay / Health Regen
 - Hazard Mitigation: Dealing with the Unexpected
 - Meteor Showers
 - Solar Flares, etc

New Possible Configurations:

- Exploration + Resource Gathering
 - Logistics - Transporting, Storing, Hauling Resources
 - Efficient resource transport
 - Increased inventory storage
 - Increased resource management efficiency (decreased costs)
 - Increased Resource Transfer Rate / Range
 - Advanced Towing Mechanics
 - Improved Resource Refinement Techniques (increased yields)
 - Excavation - Removing, Destroying, Refining Materials
 - Increased Excavation Speed
 - Reduced Digging Tool Wear / Tear
 - Ability to Detect Valuable Resources
 - Unlock Advanced Digging Techniques
 - Increased Carrying Capacity for Resources
- Exploration + Survival
 - Endurance

- Sustaining operations over long periods of deep space travel
 - Optimizing fuel consumption
 - Navigating treacherous space anomalies
 - Sustaining Crewman Health
 - Focused on long-term sustainability and efficient travel across vast distances
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 - Ex.
 - *A player with high Endurance can travel vast distances with minimal fuel expenditure, navigate complex asteroid fields, and keep their crew alive and productive during lengthy expeditions.*
- Navigation
 - Improved Fuel Efficiency
 - Enhanced Sensor Ranges / Visibility
 - Ability to Chart Routes
 - Reduced risk of collisions w/ debris
 - Improved maneuverability
- Resource Gathering + Survival
 - Engineering
 - Improved Ship Engine Performance
 - Advanced Ship Upgrades
 - Reduced Repair Costs
 - Resource Refinement
 - Emergency Repairs
 - Ability to Craft Emergency Supplies (O2 Tanks, Hull Patches, etc)
 - Enhanced Shield Tech
 - Resilience
 - Withstanding the hars realities of space exploration
 - Surviving environmental hazards
 - Maintaining Ship Health
 - Efficiently utilizing resources to ensure long-term viability
 - Focused on immediate threats and ship maintenance
 - Ex.
 - *A player with high Resilience excels at minimizing resource waste, quickly repairing ship damage, and adapting to challenging environmental conditions (extreme temperatures, radiation, etc).*